

Stem Cells and Meristems: Potential Compared

LAUNCH • Science • 40 minutes

I can compare embryonic stem cells, adult tissue stem cells and plant meristem cells using evidence about self-renewal, differentiation range and function.

Prepare before pupils arrive

Setup: Prepare a source-labelled, fictional dataset and comparison cards. Check that resources distinguish source, self-renewal, differentiation potential, laboratory outcome and clinical claim.

Safety: No biological culture is authorised. Keep discussion factual and non-diagnostic; do not describe experimental work as an available cure, and reserve structured benefit/risk/ethical evaluation for W11.

Equipment: Source-and-outcome cards, supplied fictional research table, calculator, comparison matrix and enlarged branching model.

Safety: No live cells, embryos, cultures or human tissue are used. Treat medical examples sensitively, use anonymised/fictional data and do not invite disclosure or offer treatment advice.

Fallback: Use tactile source icons, reduced-choice outcome cards, pre-entered calculations and exact pupil-directed scribing while retaining all three groups and the evidence limit.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation • Paper 1 • Topic 2 • specification point 2.8, embryonic and adult animal stem cells and plant meristems.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 3 min	Define differentiation. State one role of cell division in growth or repair.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	stem cell — unspecialised source cell self-renewal — maintain the source meristem — plant growth tissue Use a branching model: one route maintains a stem-cell population through self-renewal; another route produces cells that begin differentiation. Place embryonic stem cells on a broad animal-cell branch and adult tissue stem cells on a narrower tissue- related branch, while noting that exact potential depends on type and conditions.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Read a source-evidence table using the same headings for every group: source, self-renewal evidence, differentiated outcomes and known limitation. Calculate a simple differentiation percentage as cells with the stated marker divided by total observed cells multiplied by 100, keeping it tied to the tested conditions.	Model one complete evidence-to- explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 5 min	Route each evidence card to embryonic stem cells, adult tissue stem cells or plant meristems, then justify one comparison. Read the source and outcome evidence on one card.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 12 min	Build an evidence-led comparison of embryonic stem cells, adult tissue stem cells and plant meristems, including one percentage calculation and one bounded application claim. Record: A three-way comparison, correct use of self-renewal/differentiation, one shown percentage calculation and a claim explicitly limited to the supplied evidence.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.

Stage / total time	Teaching move	Adult support / response
Exit · 4 min	Compare embryonic and adult tissue stem cells using differentiation potential and one tissue role. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Embryonic stem cells can form many animal cell types, while many adult stem cells form a more limited tissue-related range.

Correct. Source and differentiation potential differ; adult tissues and plant meristems still contain dividing stem-cell populations.

Misconception repair

Repaired. State exactly what differentiated, under which conditions, and which clinical questions remain unanswered.

Guided checkpoint

Correct. Source and differentiation potential differ; adult tissues and plant meristems still contain dividing stem-cell populations. Compare breadth of animal-cell outcomes, then keep the plant meristem in the plant context.

Broad animal-cell range

EMBRYONIC STEM CELLS. Embryonic: the evidence supports a broad animal-cell differentiation potential under tested conditions. Look for the early-embryo source and broad animal-cell outcomes.

Early embryo source

EMBRYONIC STEM CELLS. Embryonic: source identifies this stem-cell group, not a guaranteed clinical result. Classify first by source.

Blood-cell replacement

ADULT TISSUE STEM CELLS. Adult tissue: these stem cells maintain a tissue-related range of blood cells. The source is an adult tissue and the outcomes remain tissue-related.

Skin maintenance

ADULT TISSUE STEM CELLS. Adult tissue: self-renewal and differentiation support maintenance and repair. Use the adult tissue role: ongoing replacement.

Root-tip growth

PLANT MERISTEMS. Meristem: this plant growth region supplies new root cells. The root-tip and plant-growth clues identify a meristem.

Shoot and leaf tissues

PLANT MERISTEMS. Meristem: continued division supports new plant organs and specialised tissues. Keep plant growth and animal tissue repair in separate contexts.

Plant clone potential

PLANT MERISTEMS. Meristem: the evidence supports plant tissue-culture potential under controlled conditions, not automatic growth in any setting. The outcome is a new plant produced from plant growth tissue.

Model limitation

A branching diagram makes developmental choices look fixed and equal. Real differentiation depends on signals, gene activity and conditions; the model omits failure, cell death, timing and ethical questions reserved for the next sequence.

Independent evidence

A three-way comparison, correct use of self-renewal/differentiation, one shown percentage calculation and a claim explicitly limited to the supplied evidence.

Supported: Sort six cards, complete one similarity-and-difference frame and calculate a percentage from a worked fraction. Use colour-coded source icons and the stem: Both can ___. Unlike ___, ___ can ___ under the tested conditions.

Standard: Compare all three groups, calculate one outcome percentage and explain one biological role using evidence. Use COMPARE and CALCULATE separately; then add a sentence beginning The evidence does not show ___.

Stretch: Evaluate a claimed application by combining source, potential, quantitative outcome and limitations without moving into unsupported cure language. Distinguish laboratory differentiation, tissue function, treatment safety and patient benefit.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W10L2_Stem_Cells_And_Meristems_Explore.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.